Machine tests, maximum torque, generator mode

Answers and worked solutions · Day 11 · Tue 3 Nov · pack M8

5. Practice

From your course files

Jun24 Q3b (past exam, 4 marks).

$R_1 = 2\ \Omega$. This is given per phase, so no halving is needed.

Blocked rotor, per phase: $V_1 = 80/\sqrt{3} = 46.19$ V, $I_1 = 5$ A, $P_1 = 390/3 = 130$ W.

$$R = R_1 + R_2' = \frac{130}{5^2} = 5.2\ \Omega \quad \Rightarrow \quad R_2' = 5.2 - 2 = \mathbf{3.2\ \Omega}$$

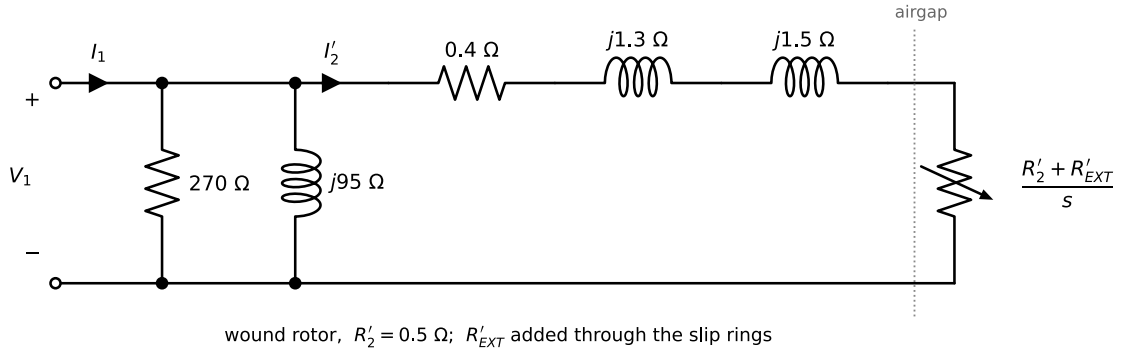
$$S_1 = 46.19 \times 5 = 230.9\ \text{VA}, \quad Q_1 = \sqrt{230.9^2 - 130^2} = 190.9\ \text{VAr}, \quad X = \frac{190.9}{5^2} = \mathbf{7.635\ \Omega}$$

So $X_1 = X_2' = 3.818\ \Omega$.

> **CHECK AGAINST THE COURSE ANSWERS:** R1 = 2 Ω, R2′ = 3.2 Ω, X = 7.635 Ω. All three agree.

The no-load test is not needed for the series parameters, but it gives $R_C = 239.6^2/90 = 637.9\ \Omega$ and $X_M = 239.6^2/833.8 = 68.85\ \Omega$ (using $P_1 = 90$ W, $S_1 = 838.6$ VA, $Q_1 = 833.8$ VAr).

Jun23 Q3b (past exam, 5 marks).



wound rotor, $R_2' = 0.5\ \Omega$; R_{EXT}' added through the slip rings

Jun23 Q3b circuit: 270 ohm parallel j95 ohm, then 0.4 ohm, j1.3 ohm, j1.5 ohm and (R2′ + REXT′)/s

(i) [2] $X = 1.3 + 1.5 = 2.8\ \Omega$. X_M and R_C are not needed, because they are at the terminals in the simplified circuit.

For maximum torque at $s = 1$:

$$\frac{R_2' + R_{EXT}'}{1} = |R_1 + jX| = \sqrt{0.4^2 + 2.8^2} = 2.828\ \Omega$$

$$R_{EXT}' = 2.828 - 0.5 = \mathbf{2.328\ \Omega}$$

> **CHECK AGAINST THE COURSE ANSWER:** R2EXT′ = 2.328 Ω. This agrees.

(ii) [3] Sketch. Draw three curves on τ vs n, all from $\tau = 0$ at $n = n_S$, all with **the same peak height**:

- $R_{EXT} = 0$: peak at $s = 0.5/2.828 = 0.177$ (close to n_S). Lowest starting torque.
- $R_{EXT} = 0.5$ pu $= 1.164\ \Omega$: peak at $s = (0.5 + 1.164)/2.828 = 0.588$. Higher starting torque.
- $R_{EXT} = 1$ pu $= 2.328\ \Omega$: peak at $s = 1$ ($n = 0$). Starting torque equals τ_{max}.

Near n_S the higher-resistance curves are less steep, because there $\tau \approx 3V_1^2 s/(\omega_S R_2')$. See the figure in Section 1.5 for the shape.

PQ 7.11 (practice questions).

a. $60(50)/1420 = 2.11$, so $p = 2$ and $n_S =$ **1500 rpm**.
b. $\omega_R = 2\pi(1420)/60 = 148.7$ rad/s, so $\tau_R = 2200/148.7 =$ **14.80 Nm**.
c. i. $P_{in} = \sqrt{3} V_L I_L \text{ pf} = \sqrt{3}(415)(4.9)(0.78) =$ **2.747 kW**. (ii) $P_{out}/\eta = 2200/0.8 =$ **2.750 kW**.
d. Ideal efficiency $= 1 - s = n_R/n_S = 1420/1500 =$ **94.67 %**. It only includes the rotor copper loss; stator copper, iron, and friction and windage losses make the actual efficiency (80 %) lower.

Course answers: 1500 rpm, 14.80 Nm, 2.747 kW, 2.750 kW, 94.67 %. (Full working is also in pack M7.)

PQ 7.12 (practice questions).

(a) Blocked rotor: $V_1 = 65/\sqrt{3} = 37.53$ V, $I_1 = 4.8$ A, $P_1 = 116.7$ W.
$R = 116.7/4.8^2 = 5.064\ \Omega$, so $R_2' = 5.064 - 3 = 2.064\ \Omega$. $S_1 = 180.1$ VA, $Q_1 = 137.2$ VAr, $X = 137.2/4.8^2 = 5.957\ \Omega$, and $X_1 = X_2' = 2.978\ \Omega$.

No load: $V_1 = 239.6$ V, $P_1 = 68.33$ W, $S_1 = 718.8$ VA, $Q_1 = 715.5$ VAr.
$R_C = 239.6^2/68.33 = 840.1\ \Omega$ and $X_M = 239.6^2/715.5 = 80.23\ \Omega$.

(b) Draw the course simplified circuit (Section 1): $840.1\ \Omega \parallel j80.23\ \Omega$ across $V_1 = 239.6$ V, then $R_1 = 3\ \Omega$, $jX_1 = jX_2' = j2.978\ \Omega$ and R_2'/s in series.

(c) $p = 2$, $n_S = 1500$ rpm, $s = 45/1500 = 0.03$, $R_2'/s = 68.79\ \Omega$, $\omega_S = 157.08$ rad/s.

$I_2' = 239.6/(3 + 68.79 + j5.957) = 3.326\angle -4.74^\circ$ A.

$I_1 = I_2' + 239.6/840.1 + 239.6/(j80.23) = 4.858\angle -42.2^\circ$ A, so pf $= 0.741$ lag.

$P_{IN} = 3(239.6)(4.858)(0.741) = 2.588$ kW.

$\tau = 3(3.326)^2(68.79)/157.08 = 14.53$ Nm, and
$P_{OUT} = 14.53 \times (2\pi \times 1455/60) = 2.215$ kW, so $\eta = 85.6\ \%$.

(d) $P_{Cu} = 3(3.326)^2(3 + 2.064) = 168.1$ W, $P_{Fe} = 3(239.6)^2/840.1 = 205.0$ W, and $\eta = 2215/(2215 + 373.1) = 85.6\ \%$. This is the same as (c), which confirms the power balance.

Course answers: $R_2' = 2.064\ \Omega$, $X = 5.956\ \Omega$, $R_C = 840.2\ \Omega$, $X_M = 80.23\ \Omega$; (c) [$s = 0.030$, $I_2' = 3.323\angle -4.738^\circ$ A] 4.855 A, 0.7408 lag, 2.585 kW, 14.52 Nm, 2.212 kW, 85.58 %; (d) Cu 167.8 W, Fe 205.0 W, total 372.8 W, 85.58 %. These differ from the working above only in the third or fourth figure (intermediate rounding). *Hint:* copper loss uses I_2' through $R_1 + R_2'$.

PQ 7.13 (practice questions).

Everything per phase. $R_1 =$ **1 Ω** (given per phase).

Blocked rotor: $V_1 = 90/\sqrt{3} = 51.96$ V, $I_1 = 8$ A, $P_1 = 90$ W.
$R_1 + R_2' = 90/8^2 = 1.406\ \Omega$, so $R_2' =$ **0.406 Ω**. $S_1 = 415.7$ VA,
$Q_1 = \sqrt{415.7^2 - 90^2} = 405.8$ VAr, $X = 405.8/8^2 =$ **6.341 Ω** ($X_1 = X_2' = 3.171\ \Omega$).

No load: $V_1 = 239.6$ V, $I_1 = 5.4$ A, $P_1 = 100$ W. $R_C = 239.6^2/100 =$ **574.1 Ω**.
$S_1 = 1293.8$ VA, $Q_1 = 1290.0$ VAr, $X_M = 239.6^2/1290.0 =$ **44.50 Ω**.

Course answers: 0.406, 6.341, 574.1, 44.50 Ω. All agree.

PQ 7.15 (practice questions).

$V_1 = 460/\sqrt{3} = 265.6$ V. 60 Hz and 1760 rpm give $p = 2$, $n_S = 1800$ rpm, $\omega_S = 188.5$ rad/s. $X = 2.3\ \Omega$ and $|R_1 + jX| = |0.25 + j2.3| = 2.314\ \Omega$.

(a) $s_{\tau max} = 0.2/2.314 = 0.08645$, so $n = 1800(1 - 0.08645) = 1644$ rpm.

(b) (i) At n_S, $s = 0$ and $\tau = 0$. (ii) $s = 40/1800 = 0.02222$, $R_2'/s = 9.0\ \Omega$:
$\tau = 3(265.6)^2(9.0)/\{188.5[(9.25)^2 + 2.3^2]\} = 111.2$ Nm. (iii) $R_2'/s = 2.314\ \Omega$:
$\tau = 218.9$ Nm. (iv) $s = 1$: $\tau = 3(265.6)^2(0.2)/\{188.5[0.45^2 + 2.3^2]\} = 40.88$ Nm.

(c) Sketch: τ = 40.88 Nm at 0 rpm, rising to the 218.9 Nm peak at 1644 rpm, then falling steeply through 111.2 Nm at 1760 rpm to 0 at 1800 rpm.

(d) The total rotor resistance is $2.314\ \Omega$, so the external part is $2.114\ \Omega$. Starting torque $= \tau_{max} = 218.9$ Nm.

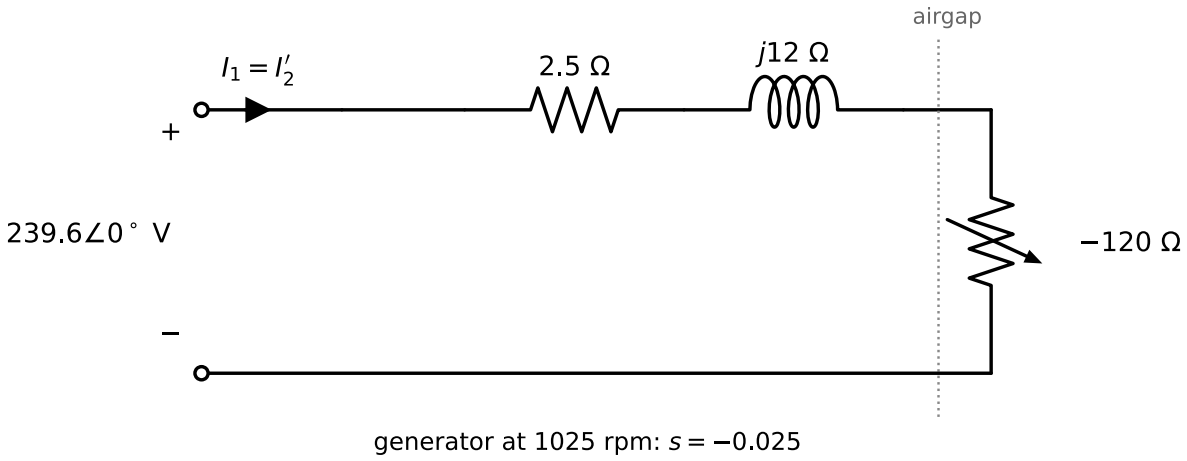
Course answers: (a) 1644 rpm (b) 0, 111.2, 218.9, 40.88 Nm (d) 2.314 Ω, 218.9 Nm. *Hint:* 1760 rpm at 60 Hz means 4 poles; in (d) the question wants the *total* rotor resistance.

Jun24 Q3c (past exam, 10 marks).

(i) [3] 960 rpm at 50 Hz: $p = 3000/960 = 3.125$, so $p = 3$ (6 poles).
$n_S = 60 \times 50/3 = \mathbf{1000}$ rpm and $\omega_S = 104.72$ rad/s.

$$s = \frac{1000 - 1025}{1000} = \mathbf{-0.025}, \qquad \frac{R_2'}{s} = \frac{3}{-0.025} = \mathbf{-120\ \Omega}$$

(ii) [5] Circuit: V_1, then $R_1 = 2.5\ \Omega$, then $jX = j12\ \Omega$, then $R_2'/s = -120\ \Omega$, all in series (no shunt branch).



Jun24 Q3c circuit: 239.6 V, 2.5 ohm, j12 ohm and R2′/s = -120 ohm in series, no shunt branch

$V_1 = 415/\sqrt{3} = 239.6\angle 0^\circ$ V.

$$Z = 2.5 - 120 + j12 = -117.5 + j12 = 118.1\angle 174.2^\circ\ \Omega$$

$$I_1 = \frac{239.6\angle 0^\circ}{118.1\angle 174.2^\circ} = \mathbf{2.029\angle -174.2^\circ}\ \text{A}$$

$$P_{IN} = 3V_1 I_1 \cos\phi = 3(239.6)(2.029)\cos(-174.2^\circ) = \mathbf{-1451}\ \text{W}$$

The negative sign means 1.451 kW flows *into* the supply. The machine is generating. ($Q_{IN} = +148$ VAr, so it still absorbs reactive power.)

(iii) [2] With no shunt branch, $I_2' = I_1$:

$$P_{AG} = 3I_2'^2\frac{R_2'}{s} = 3(2.029)^2(-120) = \mathbf{-1.481}\ \text{kW}, \qquad \tau = \frac{P_{AG}}{\omega_S} = \frac{-1481}{104.72} = \mathbf{-14.15}\ \text{Nm}$$

Power flow check: shaft input $= |P_{AG}|(1-s) = 1481 \times 1.025 = 1519$ W. Rotor copper loss $= 3(2.029)^2(3) = 37$ W, leaving $|P_{AG}| = 1481$ W. Stator copper loss $= 3(2.029)^2(2.5) = 31$ W. Electrical output $= 1481 - 31 = 1451$ W. This matches $|P_{IN}|$.

> **CHECK AGAINST THE COURSE ANSWERS:** 1000 rpm, −0.025, −120 Ω; $I_1 = 2.029\angle -174.2^\circ$ A; $P_{in} = -1451$ W; $P_{ag} = -1.482$ kW; $\tau = -14.15$ Nm. All agree. (The course's −1.482 kW and this −1.481 kW differ only in rounding: the unrounded value is −1481.5 W.)

PQ 7.14 (practice questions, if time: parts (d)–(f)).

$V_1 = 415/\sqrt{3} = 239.6$ V, $p = 1$, $n_S = 3000$ rpm, $\omega_S = 314.16$ rad/s, $X = 0.6\ \Omega$. Rated slip $s = 210/3000 = 0.07$, $R_2'/s = 2.286\ \Omega$. Parts (a)–(c) are needed for (d).

(a) $s = 1$: $I = 239.6/|0.36 + j0.6| =$ **342.4 A**; $P = 3(342.4)^2(0.36) =$ **126.6 kW**; $\tau = 3(342.4)^2(0.16)/314.16 =$ **179.2 Nm**. (If you get 93.7 A you have used rated speed, not standstill.)

(b) $I_2' = 239.6/(0.2 + 2.286 + j0.6) = 93.70\angle -13.57^\circ$ A; $I_1 = I_2' + 239.6/(j12) =$ **100.3 A**, pf = **0.9083 lag**, $P_{IN} = 3(239.6)(100.3)(0.9083) =$ **65.47 kW**.

(c) $\tau_R = 3(93.70)^2(2.286)/314.16 =$ **191.6 Nm**; $P_{OUT} = \tau_R\,\omega_S(1-s) =$ **55.99 kW**; $\eta = 55.99/65.47 =$ **85.52 %**.

(d) $\tau_{start}/\tau_R = 179.2/191.6 =$ **0.935** (official 0.9353, from rounded values). $I_{start}/I_R = 342.4/100.3 =$ **3.414** (starting current without X_M, as in (a)).

(e) $s_{\tau max} = 0.16/|0.2 + j0.6| = 0.2530$, so $n = 3000(1 - 0.2530) =$ **2241 rpm**. $\tau_{max} = \dfrac{3V_1^2}{2\omega_S(R_1 + |R_1 + jX|)} =$ **329.3 Nm**, which is $329.3/191.6 =$ **1.719** times rated torque.

(f) At no load $I_2' \approx 0$, so $I_1 = V_1/X_M = 239.6/12 =$ **19.97 A**, which is $19.97/100.3 =$ **0.1991** of rated current.

(g) Sketch: τ/τ_R starts at 0.94 at 0 rpm, rises to a peak of 1.72 at 2241 rpm, falls steeply through 1.0 at 2790 rpm to 0 at 3000 rpm.

Course answers: (a) 342.4 A, 126.6 kW, 179.2 Nm (b) 100.3 A, 0.9083 lag, 65.47 kW (c) 191.6 Nm, 55.99 kW, 85.52 % (d) 0.9353, 3.414 (e) 2241 rpm, 329.3 Nm, 1.719 (f) 19.97 A, 0.1991.

New questions

New Q1 (easy) [5 marks].

$R_1 = 2.4/2 = \mathbf{1.2\ \Omega}$ (a line-to-line reading in star is two phases in series).

Blocked rotor: $V_1 = 43.30$ V, $P_1 = 200$ W. $R = 200/81 = 2.469\ \Omega$, so $R_2' = \mathbf{1.269\ \Omega}$.
$S_1 = 389.7$ VA, $Q_1 = 334.5$ VAr, $X = 334.5/81 = 4.129\ \Omega$, so $X_1 = X_2' = \mathbf{2.065\ \Omega}$.

No load: $V_1 = 230.9$ V, $P_1 = 83.33$ W, $S_1 = 923.8$ VA, $Q_1 = 920.0$ VAr.
$R_C = 230.9^2/83.33 = \mathbf{640\ \Omega}$ and $X_M = 230.9^2/920.0 = \mathbf{57.97\ \Omega}$.

New Q2 (easy–medium) [6 marks].

$V_1 = 239.6$ V, $n_S = 1000$ rpm, $\omega_S = 104.72$ rad/s, $X = 3.6\ \Omega$, $|0.5 + j3.6| = 3.635\ \Omega$.

a. $s_{\tau max} = 0.6/3.635 = 0.1651$, so $n = 1000(1 - 0.1651) = \mathbf{834.9}$ rpm.
$\tau_{max} = 3(239.6)^2/[2(104.72)(0.5 + 3.635)] = \mathbf{198.9}$ Nm.

b. $\tau_{start} = 3(239.6)^2(0.6)/\{104.72[(1.1)^2 + 3.6^2]\} = \mathbf{69.6}$ Nm.
$I_{START} = 239.6/|1.1 + j3.6| = \mathbf{63.7}$ A.

c. $R_{2EXT}' = 3.635 - 0.6 = \mathbf{3.035\ \Omega}$. The new $\tau_{start} = \tau_{max} = \mathbf{198.9}$ Nm. The new $I_{START} = 239.6/|0.5 + 3.635 + j3.6| = \mathbf{43.7}$ A. So you get almost 3 times the starting torque with about 30 % less current.

New Q3 (medium) [5 marks].

a. $V_L = 415 \times 25/50 = \mathbf{207.5}$ V (phase 119.8 V).

New Q3 circuit at 25 Hz: 119.8 V, 0.5 ohm, j0.9 ohm, j0.9 ohm and R2′/s

b. At 25 Hz: $X = 1.8\ \Omega$, $n_S = 500$ rpm, $\omega_S = 52.36$ rad/s. $|0.5 + j1.8| = 1.868\ \Omega$, so $s_{\tau max} = 0.6/1.868 = 0.3212$ and $n = 500(1 - 0.3212) = \mathbf{339.4}$ rpm.

$\tau_{max} = 3(119.8)^2/[2(52.36)(0.5 + 1.868)] = \mathbf{173.6}$ Nm, compared with 198.9 Nm at 50 Hz.

The lecture says that τ_{max} stays constant. That is exact only if R_1 is neglected: with $R_1 = 0$ both cases give $3V_1^2/(2\omega_S X) = 228.4$ Nm. At 25 Hz, X has halved but R_1 has not, so R_1 takes a bigger share of the impedance and τ_{max} drops by 13 %.

c. The slip speed is $1000 - 976.8 = 23.2$ rpm at 50 Hz and $500 - 475.9 = 24.1$ rpm at 25 Hz. That is almost the same. The curve has slid left with nearly the same shape near

n_S. The slip as a fraction is larger at 25 Hz (0.048 vs 0.023), so the ideal efficiency is slightly lower.

New Q4 (medium) [6 marks].

a. $n = 1040 > n_S = 1000$ rpm, so it is a **generator**.
$s = (1000 - 1040)/1000 = \mathbf{-0.04}$ and $R_2'/s = 0.6/(-0.04) = \mathbf{-15\ \Omega}$.

New Q4 generator circuit: 239.6 V, 0.5 ohm, j3.6 ohm and R2′/s = -15 ohm

b. $Z = 0.5 - 15 + j3.6 = -14.5 + j3.6 = 14.94\angle 166.1^\circ\ \Omega$.
$I_1 = 239.6/14.94\angle 166.1^\circ = \mathbf{16.04\angle -166.1^\circ}$ A.

$P_{IN} = 3(239.6)(16.04)\cos(-166.1^\circ) = \mathbf{-11.19}$ kW, so 11.19 kW is delivered to the grid. $Q_{IN} = \mathbf{+2.78}$ kVAr, so the machine still absorbs reactive power from the grid.

c. $P_{AG} = 3(16.04)^2(-15) = \mathbf{-11.57}$ kW and $\tau = -11574/104.72 = \mathbf{-110.5}$ Nm. The turbine supplies $|P_{AG}|(1 - s) = 11.57 \times 1.04 = \mathbf{12.04}$ kW (check: $110.5 \times 2\pi \times 1040/60 = 12.04$ kW).

Losses: rotor Cu 463 W and stator Cu 386 W. $12.04 - 0.463 - 0.386 = 11.19$ kW, which matches (b).

New Q5 (hard, chained) [10 marks].

$V_1 = 230.9$ V, $n_S = 1500$ rpm, $\omega_S = 157.08$ rad/s. $R_1 = 1.2$, $R_2' = 1.269$, $X = 4.129\ \Omega$, $R_C = 640\ \Omega$, $X_M = 57.97\ \Omega$.

a. $s = 60/1500 = 0.04$ and $R_2'/s = 31.73\ \Omega$.

New Q5 circuit at rated speed: 230.9 V, 640 ohm parallel j57.97 ohm, then 1.2 ohm, j2.065 ohm, j2.065 ohm, 31.73 ohm

$I_2' = 230.9/(1.2 + 31.73 + j4.129) = 6.959\angle -7.15^\circ$ A.

$I_1 = I_2' + 230.9/640 + 230.9/(j57.97) = \mathbf{8.736\angle -33.7^\circ}$ A, so pf $= \mathbf{0.832}$ lag.

$P_{IN} = 3(230.9)(8.736)(0.832) = \mathbf{5.034}$ kW.

$\tau_R = 3(6.959)^2(31.73)/157.08 = \mathbf{29.34}$ Nm. $P_{OUT} = 29.34 \times 150.80 = \mathbf{4.425}$ kW, so $\eta = \mathbf{87.9\ \%}$.

b. $|1.2 + j4.129| = 4.300\ \Omega$, so $s_{\tau max} = 1.269/4.300 = 0.2951$ and $n = \mathbf{1057}$ rpm.
$\tau_{max} = 3(230.9)^2/[2(157.08)(1.2 + 4.300)] = \mathbf{92.6}$ Nm, and $\tau_{max}/\tau_R = \mathbf{3.16}$.

$\tau_{start} = 3(230.9)^2(1.269)/\{157.08[(2.469)^2 + 4.129^2]\} = 55.8$ Nm, so $\tau_{start}/\tau_R = \mathbf{1.90}$.

c. $s = -0.02$ and $R_2'/s = -63.46\ \Omega$. The torque formula gives $\tau = \mathbf{-16.6}$ Nm. The machine is generating: the shaft must be driven against 16.6 Nm.

d. $I_{NL} = |230.9/640 + 230.9/(j57.97)| = 4.00$ A. This matches the test, as it should. $4.00/8.736 = \mathbf{0.46}$ of rated current. That is typical: the lecture gives 25–60 %, which is mostly magnetising current needed because of the airgap.

New Q6 (short answer, words only) [6 marks].

a. At no load the rotor runs at almost synchronous speed, so very little rotor current flows. The stator must still produce the rotating magnetic field. Because the flux has to cross an airgap, this needs a large magnetising current, typically a quarter to over half of rated current. That current is modelled by the magnetising reactance. A smaller in-phase current supplies the hysteresis and eddy-current losses in the stator iron, which are modelled by the core-loss resistance. A further small in-phase current supplies the friction and windage losses, because the rotor is still turning. The current is mostly magnetising current, which lags the voltage by nearly 90 degrees, so the power factor is very low.

b. There would be no rotor copper loss, so efficiency would rise. The slip at maximum torque is proportional to rotor resistance, so it would be zero. The torque–speed curve near synchronous speed would become vertical, like a synchronous machine, and the speed would not drop with load. The starting torque is proportional to rotor resistance, so it would be zero, and the motor could not start by itself.

6. Quick recall check

Question

1. The magnetising branch ($R_C \parallel jX_M$). At low voltage it carries little current compared with the low-impedance series branch. You need R_1 from the DC test to get $R_2' = R - R_1$.
2. $s_{\tau max} = R_2'/\sqrt{R_1^2 + (X_1 + X_2')^2}$. τ_{max} is unchanged. Only $s_{\tau max}$ doubles.
3. $R_{2EXT}' = \sqrt{R_1^2 + X^2} - R_2'$.
4. $n_S = 1500$ rpm and $s = -0.04$. $s < 0$, $R_2'/s < 0$, $\tau < 0$, $P_{IN} < 0$ (generating), and $Q_{IN} > 0$ (still absorbs VArs).
5. V_1, X and ω_S all halve. R_1 and R_2' stay the same.